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Options and Futures

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Fall Term 2025

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Derivatives: Basics

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Let's start by defining some basic terminology and concepts. We start with a definition.

Definition

A derivative security is a security for which the payoff is linked to (or derived from) the value of an underlying asset.

Given the above, it's obvious that these assets are called derivatives as their payoffs are derived from the prices of other assets.

To make things more concrete we need to specify the underlying asset and to be able to deduce how the derivative payoff relates to the underlying price.

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Below is a list of some of the most common, also known as 'vanilla', derivatives. In our later analysis we will focus on the members of this list. Beside each item in the list is a brief description of its payoff profile.

- Forward and futures contracts: the obligation to buy/sell the underlying asset at a pre-specified price and on a pre-specified date.
- Call options: the right to buy the underlying asset for a pre-specified price and on (or before) a pre-specified date.
- Put options: the right to sell the underlying asset for a pre-specified price and on (or before) a pre-specified date.
- Swaps: the exchange of cashflows for a pre-specified period of time. The cashflows are determined by some preset rule and are based on the value of some underlying assets.

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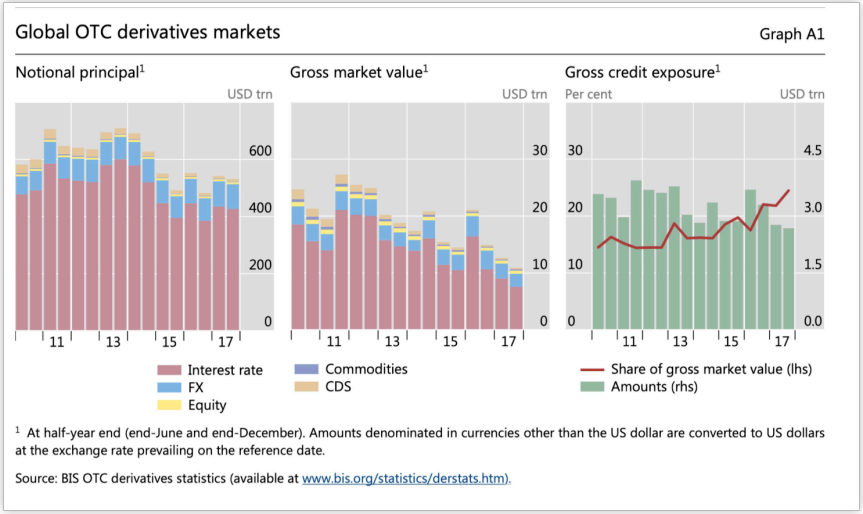
Underlyings: the reference asset that determines the derivative's payoff. Amongst other things, one can create derivatives on:

- Equities: single stocks (e.g. AMAZN or MSFT) or equity indices (e.g. the FTSE-100, S&P-500, Nikkei 225).
- Currencies: exchange rates such as USD/EUR or JPY/GBP.
- Bonds and Interest rates: for example, LIBOR, Bunds and Gilts.
- Commodities: such as Gold, Oil and Coffee.

Derivatives pricing theory studies the relationship between the price of a derivative and that of the underlying, based on the principle of no arbitrage.

Size of global OTC derivatives

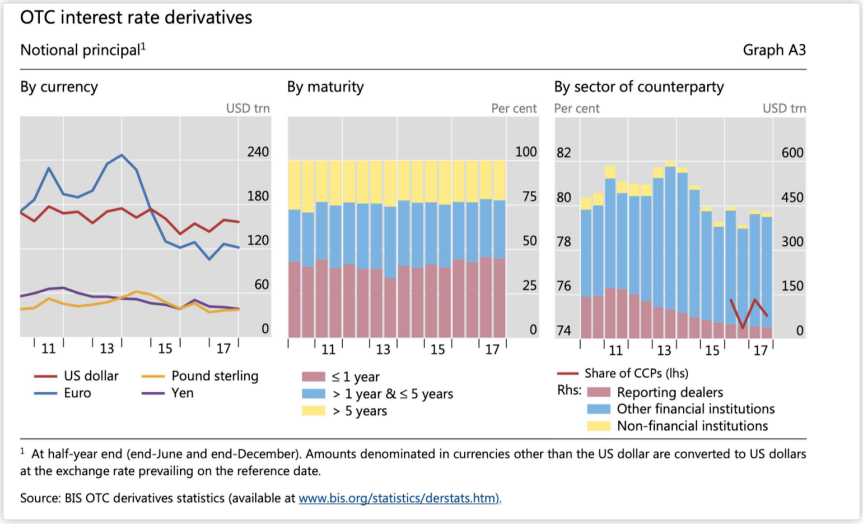
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OTC interest rate derivatives

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Finally, it is useful at this point to ask exactly what financial market participants use derivatives for. Derivatives are used for a number of reasons with the most common below;

1. Hedging: many market participants, for example corporate treasurers in oil firms or exporting firms, use derivatives to remove or reduce the risks associated with their economic activities or investment portfolios.
2. Making bets: one can use derivatives to deliberately gain exposure to certain risks e.g. if you have a view on the future value of the FTSE-100, one could exploit this using FTSE futures.
3. Create arbitrage portfolios: as derivatives are based on the prices of underlying assets, one can sometimes construct portfolios of derivatives and the underlying that yield risk-free arbitrage profits.

Forwards

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Forwards are amongst the oldest derivative contracts, having been traded (particularly in the agricultural sector) for hundreds of years.

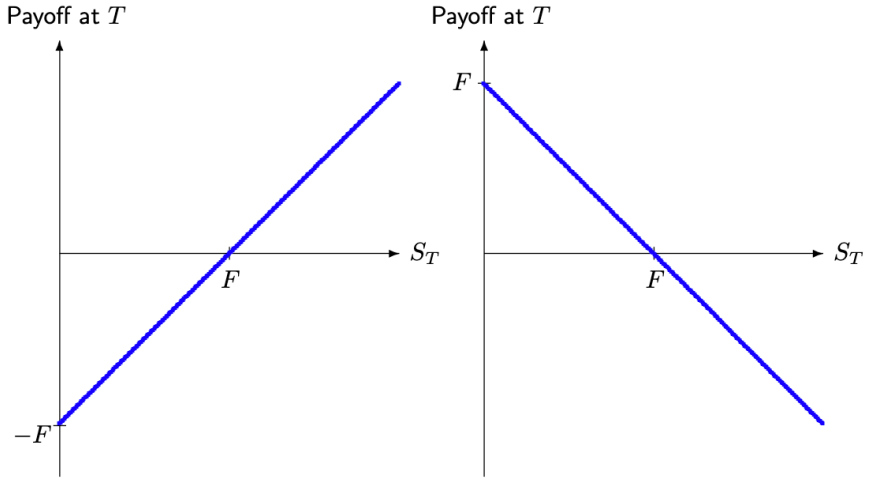
Forward contract: the obligation to buy/sell the underlying asset at a pre-specified price and on a pre-specified date

- Specifies an expiration date T
- Specifies a forward price F for a particular asset
- Long position: obligation to buy the asset payoff $S_T - F$
- Short position: obligation to sell the asset payoff $F - S_T$

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Payoff of a long/short position in a forward contract

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Pricing by absence of arbitrage

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All of our derivatives pricing, whether in discrete or continuous time, will use arbitrage pricing techniques. Let us first redefine what we mean by arbitrage.

Definition

An arbitrage opportunity is said to exist when one can construct one of the following:

- A portfolio with zero set-up cost (i.e. a zero price portfolio) but a chance of positive subsequent payoffs (and no chance of a negative payoff).
- A portfolio with a negative set-up cost and zero payoffs thereafter.

Implications of absence of arbitrage

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A first key implication of absence of arbitrage is as follows:

The law of one price: if two portfolios have identical payoffs in all states of nature, they must have the same price.

A second implication is:

The law of payoff dominance: if portfolio A guarantees a payoff at least as great as portfolio B in all states of nature, then portfolio A must command a greater price than portfolio B.

Pricing forwards

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Keys:

- We know that for a forward contract no cashflows are exchanged at the date on which the contract is struck.
- no arbitrage

Example:

	Forward	Replicating p/f
Cost at t_0	0	$S - \frac{F}{(1+r_T)^T}$
Payoff at $t_0 + T$	$S_T - F$	$S_T - F$

Thus, by absence of arbitrage, the cost of the replicating portfolio must be zero. Hence;

$$S - \frac{F}{(1 + r_T)^T} = 0$$

Pricing forwards (cont'd)

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We can extend our analysis to deal with forwards on assets that will pay a known dividend during the forward's lifetime.

Denote the present value of this dividend payment by I. Then the no-arbitrage T-period forward price on the dividend paying asset is:

$$F = (S - I)(1 + r_T)^T \tag{10}$$

Generalisations;

- When there are multiple dividend payments, I is then the present value of the entire set of dividend payments
- When dividend payments are uncertain, I should be interpreted as the present value of the expected dividend payments.

Pricing forwards (cont'd)

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With continuous compounding, at rate r, we get the expected expression for the no-arbitrage delivery price in the T period forward;

$$F = Se^{rT}$$

Stock index: can be thought of as paying a continuous dividend yield. If that continuous dividend yield is d then, using the formula above, the stock index future has delivery price;

$$F = Se^{(r-d)T}$$

Currency forwards: buying a foreign currency is buying an asset with a continuous yield, where that yield is the foreign interest rate. If we denote the foreign interest rate by r_f, we can apply our formula to price a currency forward;

$$F = Se^{(r-r_f)T}$$

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Example: Carry Trade

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The spot price of the British pound is currently \$1.6, meaning that you can exchange 1 British pound for 1.60 US dollars. The risk-free interest rate on 1-year government bonds is 0.75% in the United States and 1% in the United Kingdom. For this question assume you are a US based investor.

1. What must be the forward price of the pound for delivery 1 year from now?
2. How could an investor make risk-free arbitrage profits if the forward price was \$1.6?

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Options: The right to buy/sell the underlying asset for a pre-specified price and on (or before) a pre-specified date.

- Specifies an expiration date T.
- Specifies a strike price K for a particular asset
- Call: right to buy the asset.
- Put: right to sell the asset
- European option: can be exercised only at expiration
- American option: can be exercised any time up to expiration

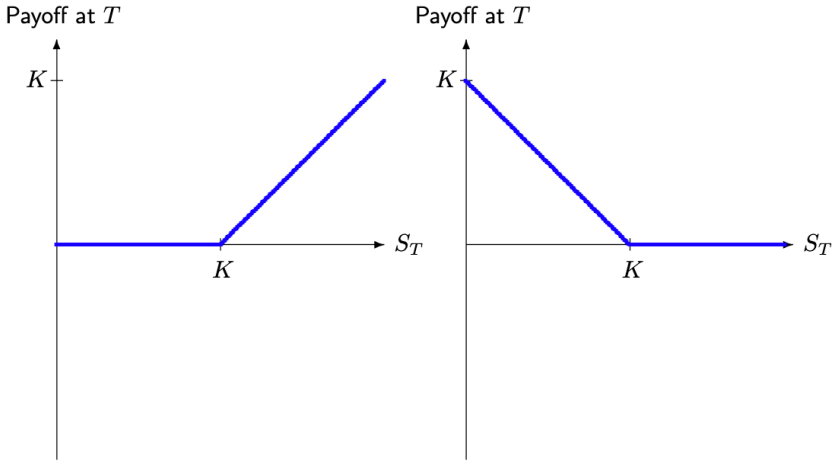
Payoffs:

- European call: $C_T = [S_T - K]^+ := max(S_T - K, 0)$
- European put: $P_T = [K - S_T]^+ := max(K - S_T, 0)$

Buying an option $\rightarrow$ right to buy/sell.
Writing an option $\rightarrow$ obligation to sell/buy.

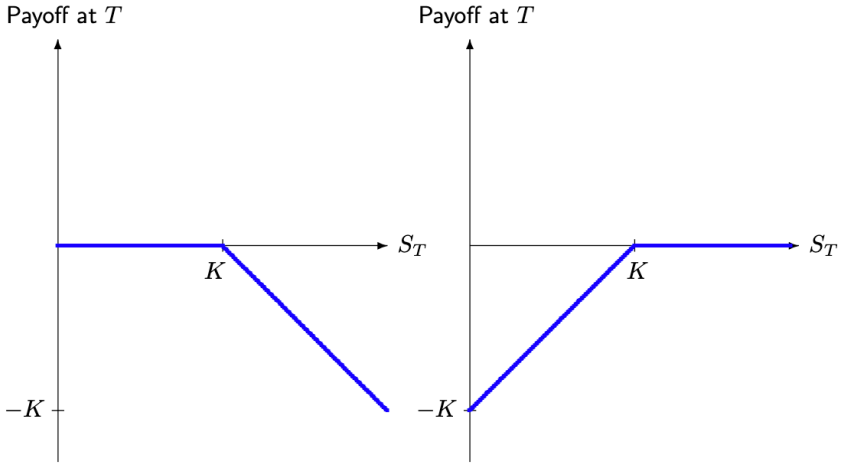
Long positions in a call and a put

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Short positions in a call and a put

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Option combinations

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- Bull call spread: combine a long position in a call option with a low strike (K_1) with a short position in a call option with a higher strike (K_2).
- Bear call spread: combine a short position in a call option with a low strike (K_1) with a long position in a call option with a higher strike (K_2).
- Butterfly call spread: go long one call with a low exercise price (K_1), short two calls with a medium strike (K_2) and long one call with a high exercise price (K_3).
- ...
- Options can be used to replicate riskless payoff. (Breeden and Litzenberger, 1978)

Bull spreads

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Consider an investor with a mildly bullish view but who wants a portfolio that's protected against extreme price moves. He can choose one of the following option combinations.

- Bull call spread: combine a long position in a call option with a low strike (K_1) with a short position in a call option with a higher strike (K_2).
- Bull put spread: combine a long position in a put option with a low strike (K_1) with a short position in a put option with a higher strike (K_2).

Both of these positions yield the desired portfolio payoff profile. A explanation for the payoff profile of the bull call spread is given below.

Bull spreads (cont'd)

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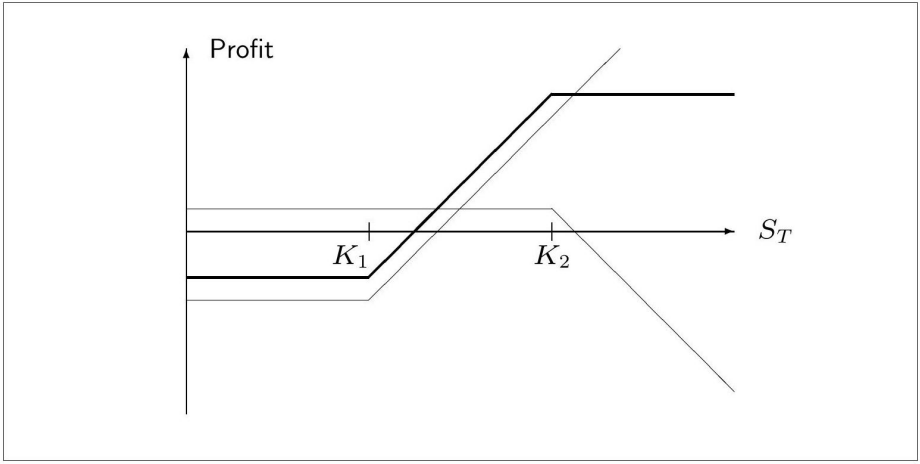
Example: payoff profile for bull call spread. Define the underlying price at maturity to be S_T. The bull call spread consists of a long call position with strike K_1 and a short call position with strike K_2 where $K_2 > K_1$;

Position	Pay-Off		
	$S_T \leq K_1$	$K_1 \leq S_T \leq K_2$	$K_2 \leq S_T$
Long 1 Call K_1	0	$S_T - K_1$	$S_T - K_1$
Short 1 Call K_2	0	0	$K_2 - S_T$
Total	0	$S_T - K_1$	$K_2 - K_1$

Self-assessment exercise: demonstrate that the bull put spread defined previously gives a similar portfolio payoff profile.

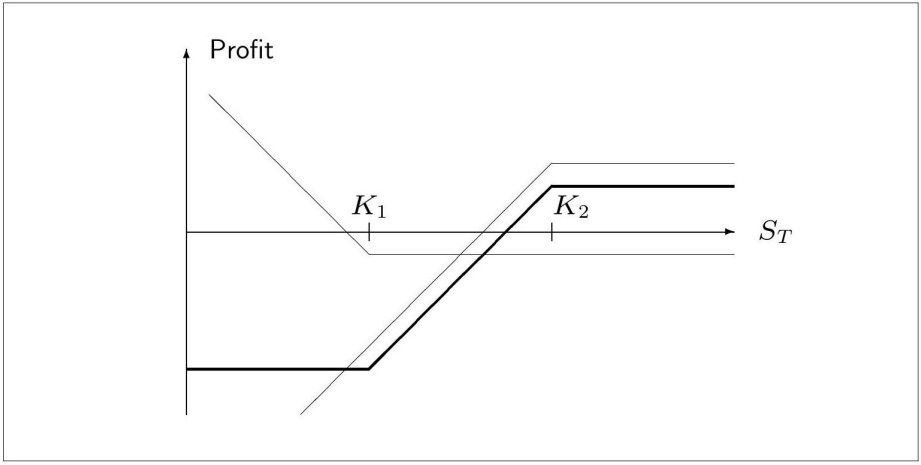
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Bull spread using Puts

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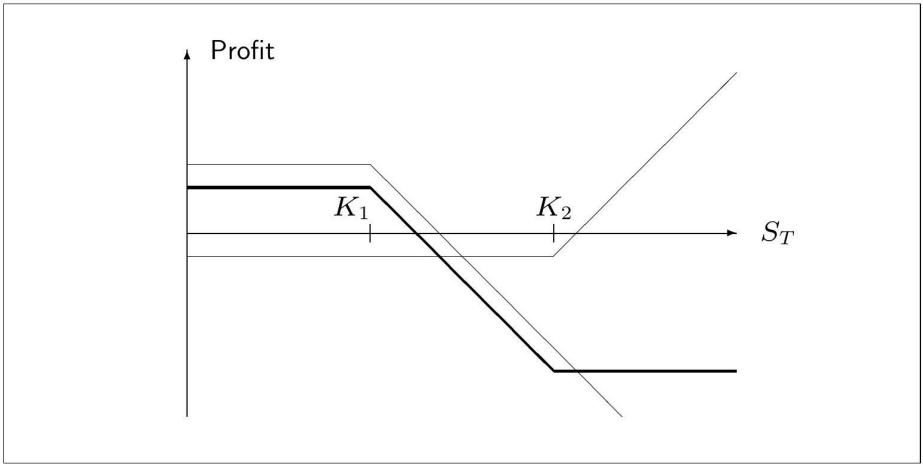
An investor who holds a mildly bearish view about the price of the underlying but wants a position that's not too sensitive to extreme market movements might construct a bear spread.

- Bear call spread: combine a short position in a call option with a low strike (K_1) with a long position in a call option with a higher strike (K_2).
- Bear put spread: combine a short position in a put option with a low strike (K_1) with a long position in a put option with a higher strike (K_2).

Self-assessment exercise: demonstrate that the bear spread positions have the desired payoff profiles.

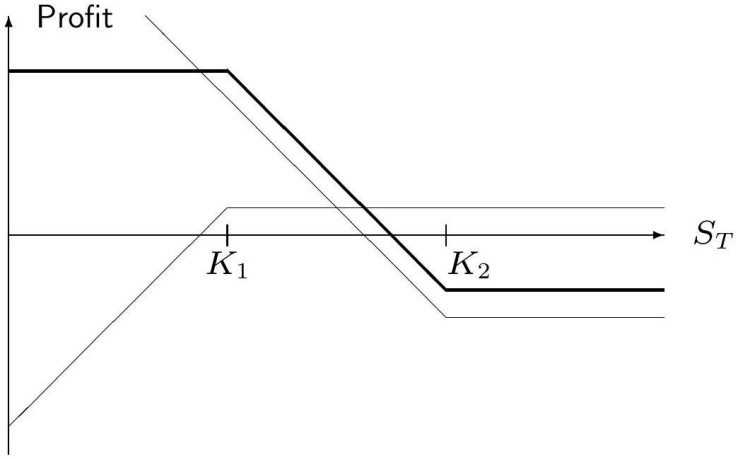
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Consider an investor who believes that volatility in the underlying will be low until maturity (i.e. the price of the underlying will change little from it's current value, either up or down).

- Butterfly spread using calls: go long one call with a low exercise price (K_1), short two calls with a medium strike (K_2) and long one call with a high exercise price (K_3).
- Butterfly spread using puts: go long one put with a low exercise price (K_1), short two puts with a medium strike (K_2) and long one put with a high exercise price (K_3).

Again, the worked example for the call spread is shown below.

Example: Payoff derivation for call Butterfly Spread

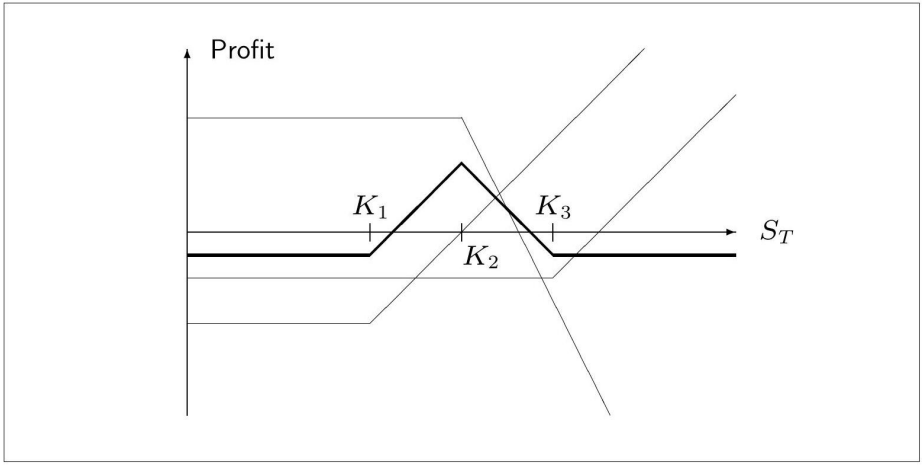
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Position	Pay-Off			
	$S_T \leq K_1$	$K_1 \leq S_T \leq K_2$	$K_2 \leq S_T \leq K_3$	$K_3 \leq S_T$
Long 1 Call K_1	0	$S_T - K_1$	$S_T - K_1$	$S_T - K_1$
Short 2 Calls K_2	0	0	$2(K_2 - S_T)$	$2(K_2 - S_T)$
Long 1 Call K_3	0	0	0	$S_T - K_3$
Total	0	$S_T - K_1$	$K_3 - S_T$	0

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Butterfly spread using Calls

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- Unless mentioned otherwise, we will assume that all options are European, and the underlying pays no dividends during the life of the option.
- Notation:
  - S: current price of the underlying.
  - C: current call price.
  - P: current put price.
- Moneyness:
  - A call option is in the money (ITM) if $S > K$.
  - A call option is out the money (OTM) if $S < K$.
  - A put option is ITM iff the corresponding call is OTM.
  - A call/put option is at the money (ATM) if $S = K$.

No-arbitrage bounds for option prices

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Assume that the underlying pays no dividends until expiration. Then we have the following upper and lower bounds on the prices of European call and put options:

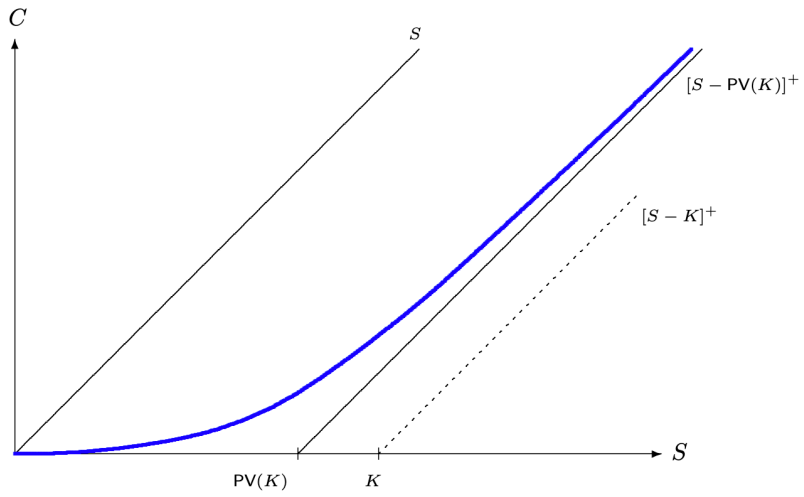
$$[S - PV(K)]^+ \leq C \leq S$$
$$[PV(K) - S]^+ \leq P \leq PV(K)$$

These bounds follow from static no-arbitrage arguments. If any of the pricing bounds is violated, there is an arbitrage opportunity:

- if $C > S$, write call and buy underlying
- if $P > PV(K)$, write put and save $PV(K)$.
- if $S - C - PV(K) > 0$, short underlying, buy call and save (or lend) $PV(K)$,
- if $PV(K) - S - P > 0$, borrow $PV(K)$, buy underlying and buy put.

Example: Black-Scholes call price

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- Lower bound on call price: $C \geq [S - PV(K)]^+$. This is the value of the option if you must decide today whether to exercise the option at expiration or not.
- If $S - PV(K) > 0$:
  - Commit to exercising
  - Sell underlying asset short, lend the amount PV(K)
  - Positive payoff $S - PV(K)$ today
  - At expiration, use the proceeds from the loan to acquire the asset at the strike price K, and close the short position.
- If $S - PV(K) \leq 0$: commit to not exercising.
- $[S - PV(K)]^+$: intrinsic value.
- $C - [S - PV(K)]^+$: time value.

Put-call parity

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Consider a European call and put on the same underlying asset, with the same strike price K, and the same maturity T.

Put-Call Parity

Suppose the underlying pays no dividends. Then, the price of the call and the put are related by:

$$C = P + S - PV(K)$$

Proof:

- Notice that: $[S_T - K]^+ - [K - S_T]^+ = S_T - K$
- Thus, going long the call and short the put gives us the linear payoff $S_T - K$, which can be replicated by buying the stock and borrowing $PV(K)$.
- The two portfolios must have the same cost, i.e. $C = P + S - PV(K)$

Q: two at the money options, call and put, which one has higher price?

Put-call parity with lumpsum dividends

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Now suppose the stock pays known dividends between time 0 and T whose present value is I.

- It is still the case that going long in the call and short in the put generates the linear payoff $S_T - K$ at the maturity date T.
- The same payoff can be obtained by buying the stock, borrowing the present value of K, and borrowing I (repaying the latter when the dividend arrives).
- The two portfolios must have the same cost:

$$C - P = S - PV(K) - I$$

Put-call parity with a continuous dividend yield

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Suppose r is the continuously compounded (annual) interest rate. Then put-call parity without dividends can be written as:

$$C_t = P_t + S_t - e^{-r(T-t)}K$$

Some types of underlying, such as stock indices and foreign currencies, pay dividends that are better approximated by a continuous dividend yield. Suppose the continuously compounded dividend rate is a constant q.

- For each share of stock at time t, the dividend over the next "instant" is qS_t.

The put-call parity formula becomes[1]:

$$C_t = P_t + e^{-q(T-t)}S_t - e^{-r(T-t)}K$$

[1] We will have a closer look at continuous dividend yields later, after we have developed continuous-time techniques.

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- Appendix A contains proofs of some results
- This material will not be examined

$\mathbb{E}[(X-K)^+]$ for X lognormal

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- To derive the Black-Scholes formula, we used the following result: Suppose X is lognormally distributed, and $\text{Var}(\log X) = \sigma^2(T-t)$. Then

$$\mathbb{E}[(X-K)^+] = \mathbb{E}(X)\Phi(d_1) - K\Phi(d_2),$$

where Φ is the standard normal cdf and

$$d_1 : \frac{\log[\mathbb{E}(X)/K] + \sigma^2(T-t)/2}{\sigma\sqrt{T-t}}, \quad d_2 := d_1 - \sigma\sqrt{T-t}$$

- To simplify notation, let $\mathbb{E}(\log X) = \alpha$, and $\text{Var}(\log X) = \beta^2$ (i.e. $\beta := \sigma\sqrt{T-t}$)

Proof

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- We can write $\log X = \alpha + \beta z$, where z is standard normal. Thus $X = e^{\alpha+\beta z}$ and $\mathbb{E}(X) = e^{\alpha+\beta^2/2}$. Moreover, $X \geq K$ is equivalent to $\log X \geq \log K$,or

$$\begin{aligned} z &\geq \frac{\log K - \alpha}{\beta} \\ &= \frac{\log K - \log \mathbb{E}(X) + \beta^2/2}{\beta} \\ &= -\frac{\log[\mathbb{E}(X)/K] - \beta^2/2}{\beta} \\ &= -d_2. \end{aligned}$$

- Recall the standard normal density: $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$
- Let Φ be the standard normal cdf.

Proof(cont'd)

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Thus we have:

$$\begin{aligned}
\mathbb{E}[(X-K)^+] &= \int_{-d_2}^{\infty} (e^{\alpha+\beta z} - K)\phi(z)dz \\
&= \int_{-d_2}^{\infty} e^{\alpha+\beta z}\phi(z)dz - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \int_{-d_2}^{\infty} \mathbb{E}(X)e^{-\frac{1}{2}\beta^2+\beta z}\cdot\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}z^2}dz - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\int_{-d_2}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(z-\beta)^2}dz - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\int_{-d_2-\beta}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(z-\beta)^2}d(z-\beta) - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\int_{-d_1}^{\infty}\phi(z-\beta)d(z-\beta) - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\Phi(d_1) - K\Phi(d_2).
\end{aligned}$$

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Breeden-Litzenberger

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- Let $\varphi(\cdot)$ be the density of S_T under the risk-neutral probability Q The price of a call option at $t = 0$ as a function of the strike price K can be written as:

$$C(K) = e^{-rT}\mathbb{E}^Q[(S_T - K)^+] = e^{-rT}\int_K^\infty (x - K)\varphi(x)dx$$

- Result [Breeden-Litzenberger (1978)]:

$$\varphi(K) = e^{rT}C''(K).$$

Proof of Breeden-Litzenberger result

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Proof: We can write

$$e^{rT}C(K) = \int_K^\infty x\varphi(x)dx - K\int_K^\infty \varphi(x)dx.$$

Differentiating:

$$\begin{aligned} e^{rT}C'(K) &= \frac{d}{dK}\int_K^\infty x\varphi(x)dx - K\frac{d}{dK}\int_K^\infty \varphi(x)dx - \int_K^\infty \varphi \\ &= -K\varphi(K) - K[-\varphi(K)] - \int_K^\infty \varphi(x)dx \\ &= -\int_K^\infty \varphi(x)dx. \end{aligned}$$

Differentiating again:

$$e^{rT}C''(K) = -[-\varphi(K)] = \varphi(K).$$

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- Put-call parity (non-dividend paying stock):

$$C_t = P_t + S_t - e^{-r(T-t)}K$$

- Lognormal property of GBM:
  Suppose $dS_t = \mu S_t dt + \sigma S_t dW_t$. Then the conditional distribution of $\log S_T$ at time t is normal:

$$\log S_T \sim \mathcal{N}\left[\log S_t + \left(\mu - \sigma^2/2\right)(T-t), \sigma^2(T-t)\right]$$

- Black-Scholes formula:
  The price of a European call option at $t < T$ is given by

$$C(S,t) = S\Phi(d_1) - e^{-r(T-t)}K\Phi(d_2)$$

  where $\Phi(\cdot)$ is the standard normal cdf, and

$$d_1 = \frac{\log(S/K) + \left(r + \sigma^2/2\right)(T-t)}{\sigma\sqrt{T-t}}, \quad d_2 = d_1 - \sigma\sqrt{T-t}$$

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- Black-Scholes PDE:
  Consider a derivative on a stock with payoff $g(S_T)$ at time T, whose price at time $t \leq T$ is given by $F(S,t)$. The Black-Scholes PDE is

$$F_t(S,t) + rSF_S(S,t) + \frac{1}{2}\sigma^2 S^2 F_{SS}(S,t) - rF(S,t) = 0$$

- Standard normal density:

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

- Expectation of the exponential of a normal random variable:
  If X normally distributed, then

$$\mathbb{E}[\exp(X)] = \exp\left[\mathbb{E}(X) + \frac{1}{2}\operatorname{Var}(X)\right]$$

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